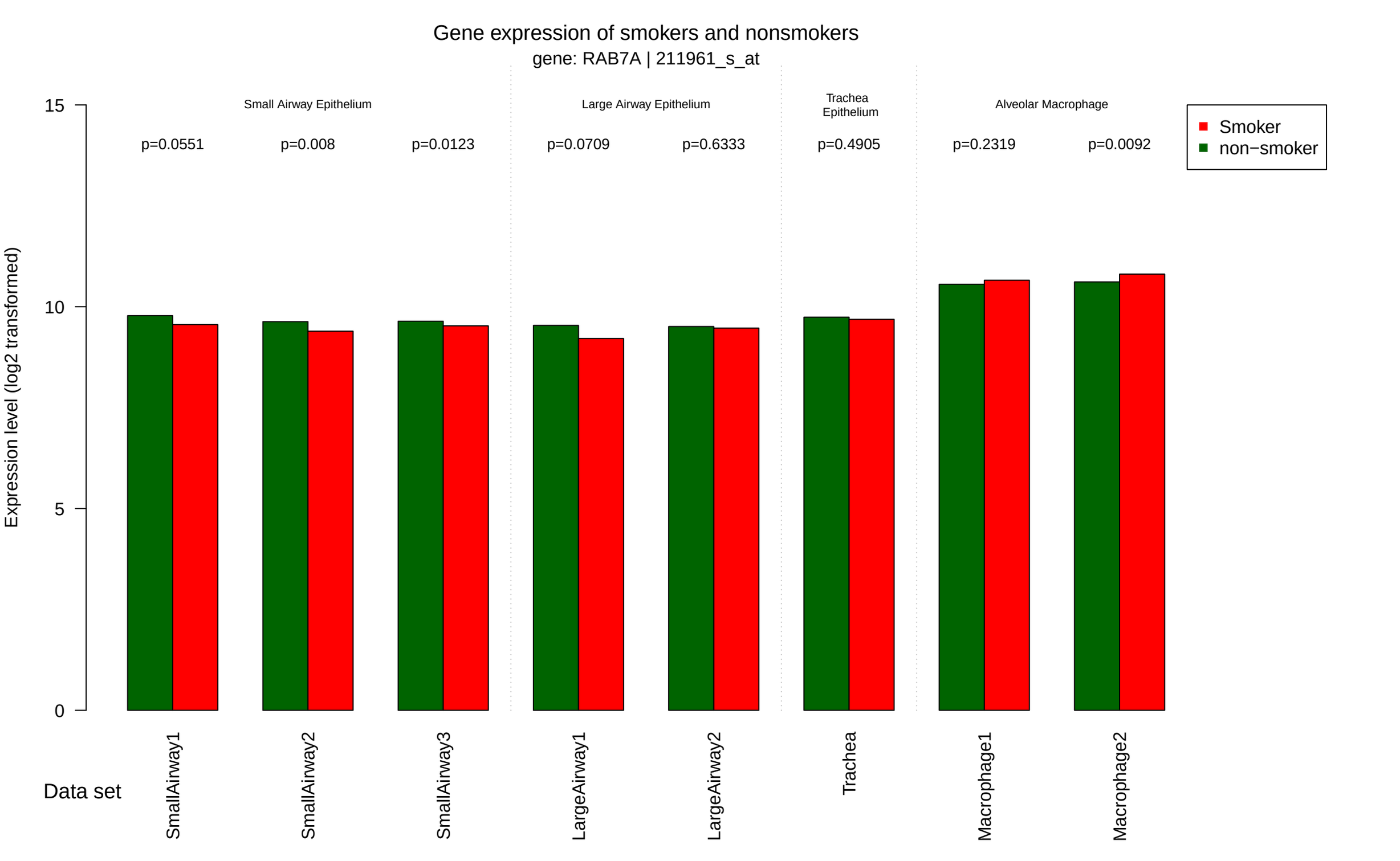
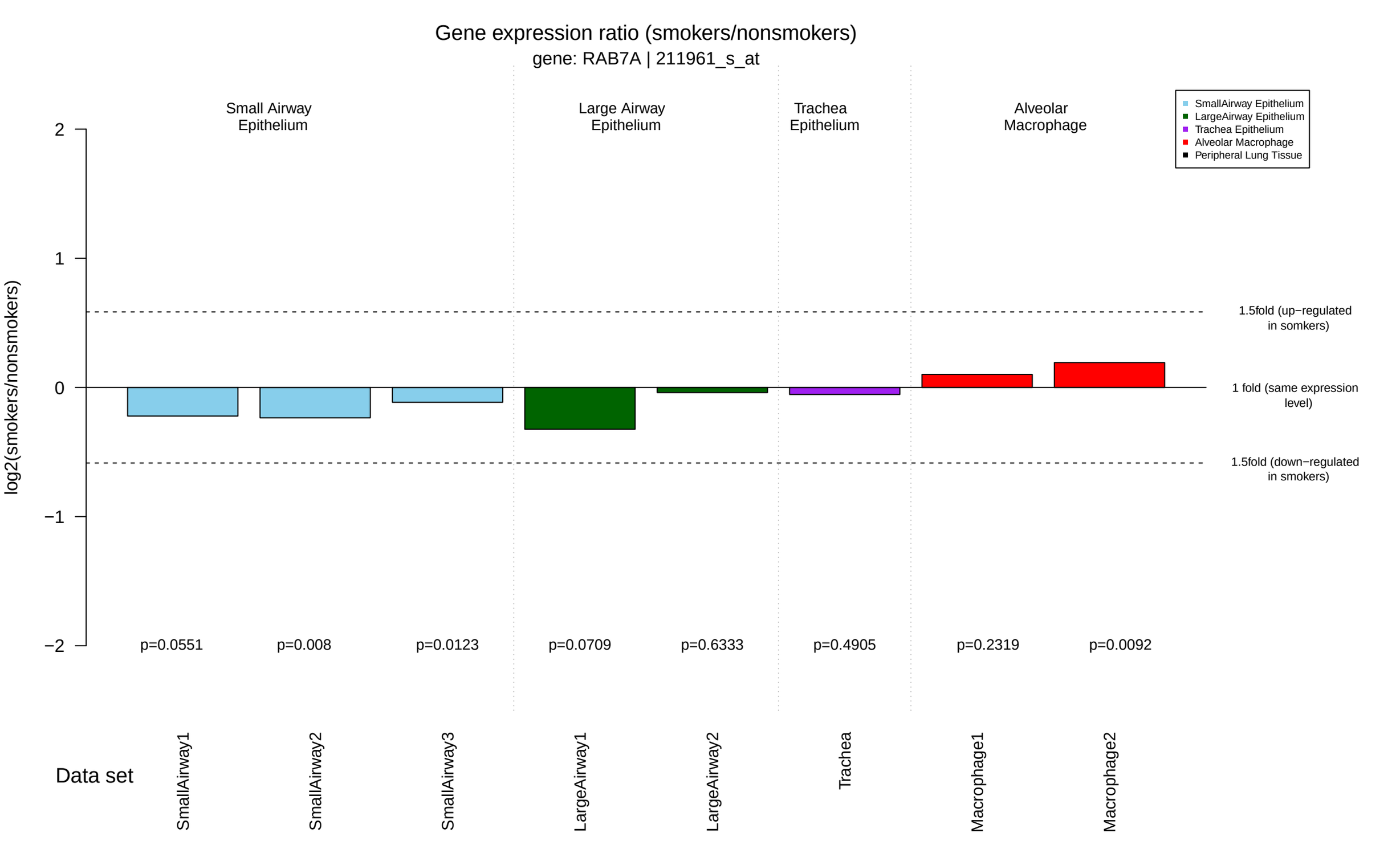
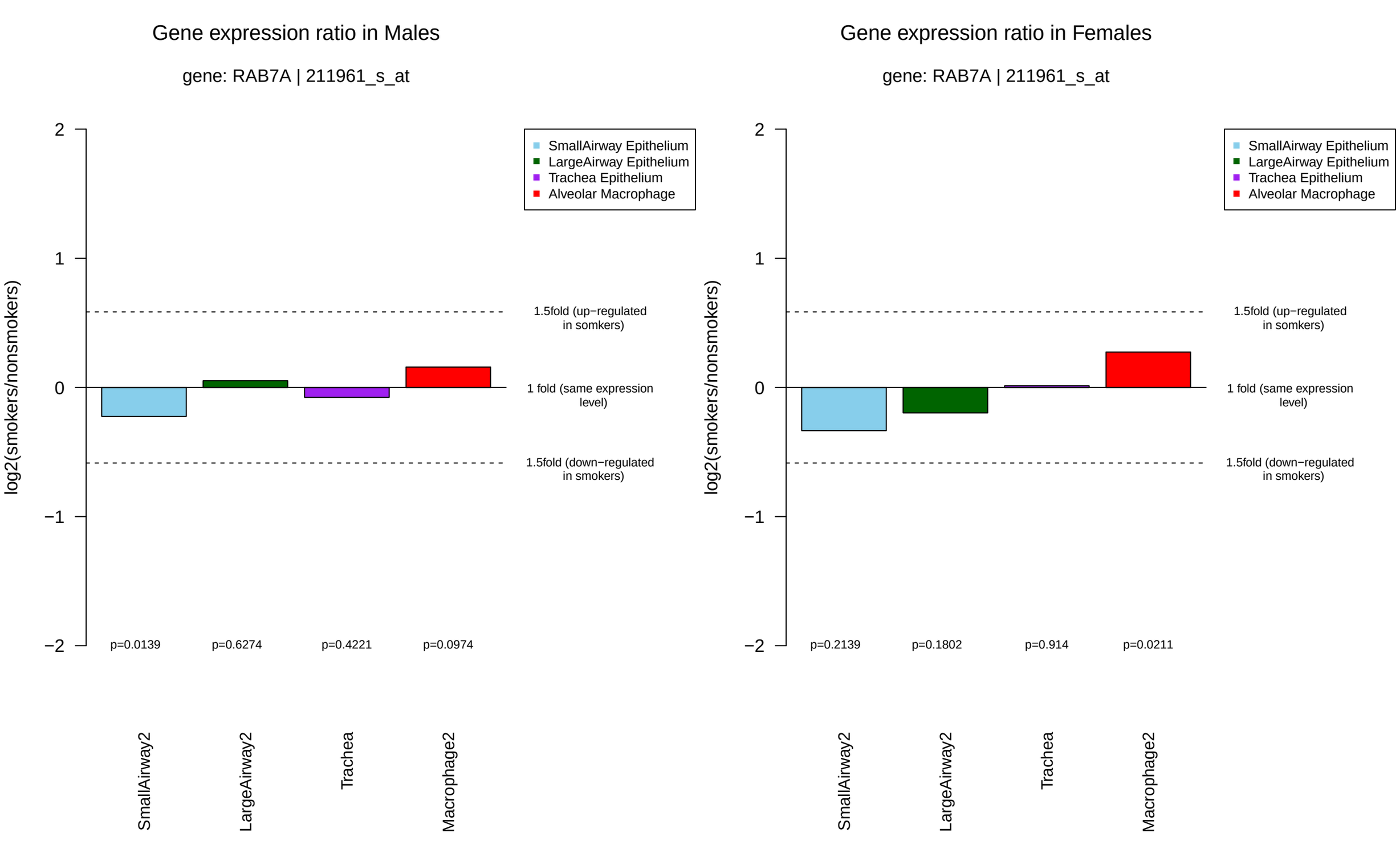
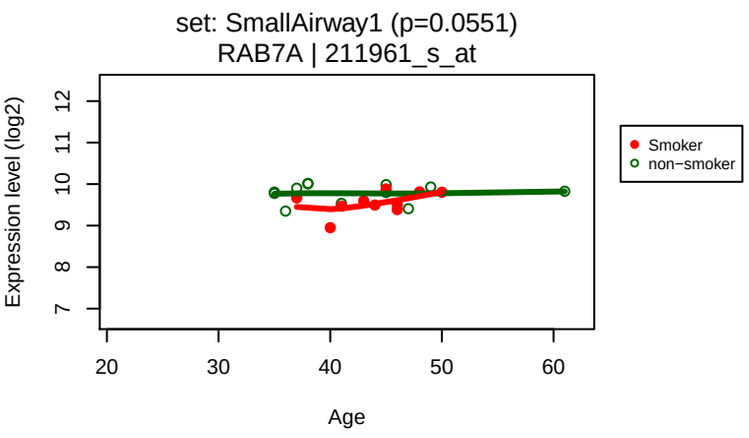
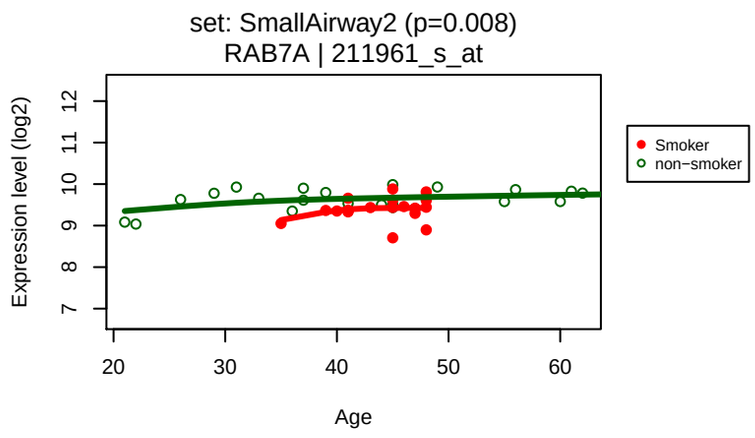
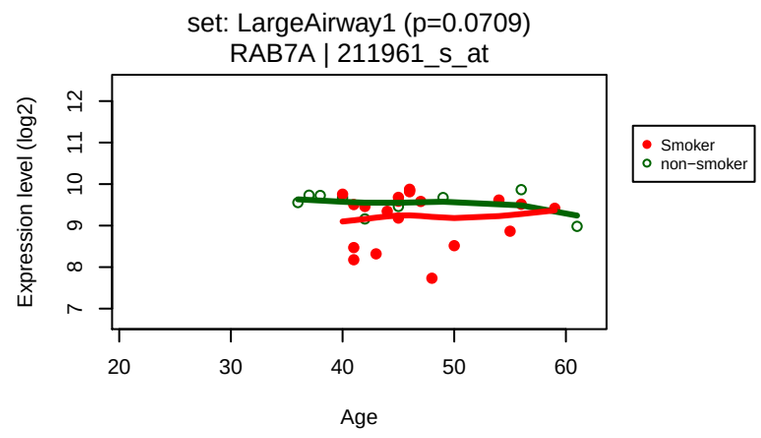

Gene expression of smokers and nonsmokers
gene: RAB7A | 211961_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0551
p=0.008
p=0.0123
p=0.0709
p=0.6333
p=0.4905
p=0.2319
p=0.0092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

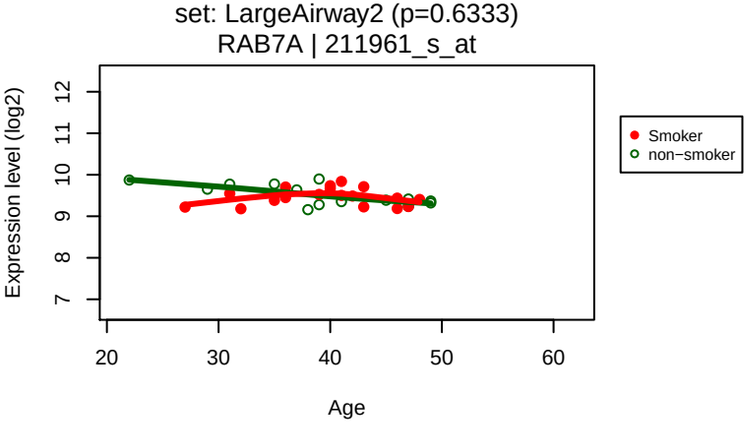

Gene expression ratio (smokers/nonsmokers)
gene: RAB7A | 211961_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0551
p=0.008
p=0.0123
p=0.0709
p=0.6333
p=0.4905
p=0.2319
p=0.0092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

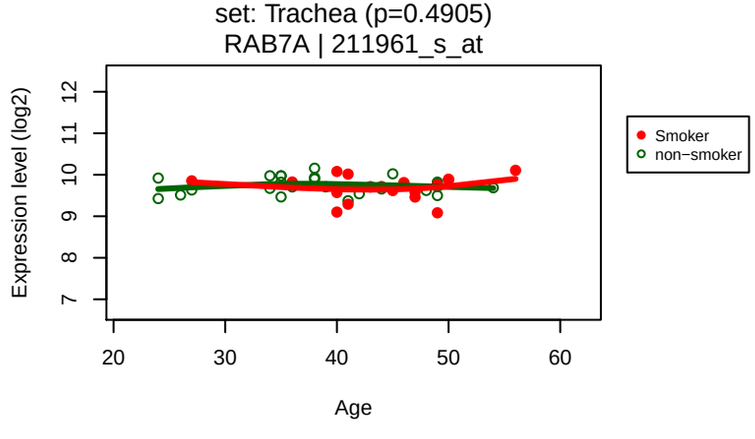

Gene expression ratio in Males
gene: RAB7A | 211961_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0139
p=0.6274
p=0.4221
p=0.0974
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB7A | 211961_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2139
p=0.1802
p=0.914
p=0.0211
SmallAirway2
LargeAirway2
Trachea
Macrophage2

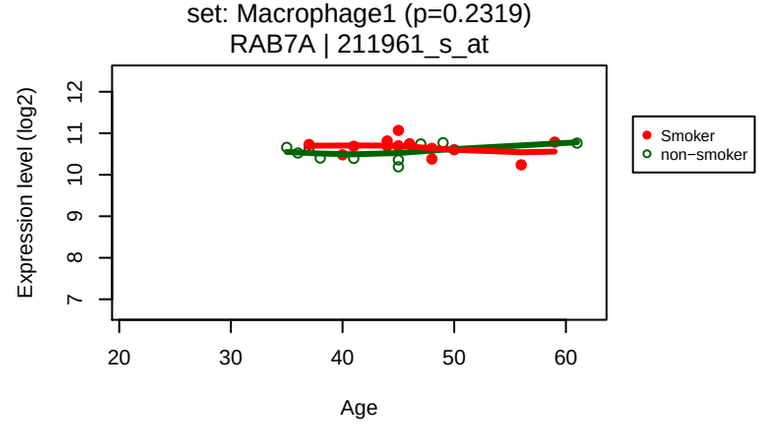

set: SmallAirway1 (p=0.0551)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker

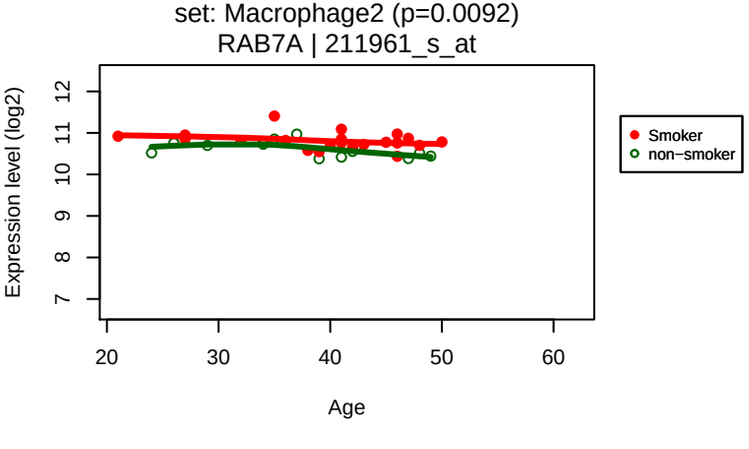

set: SmallAirway2 (p=0.008)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker

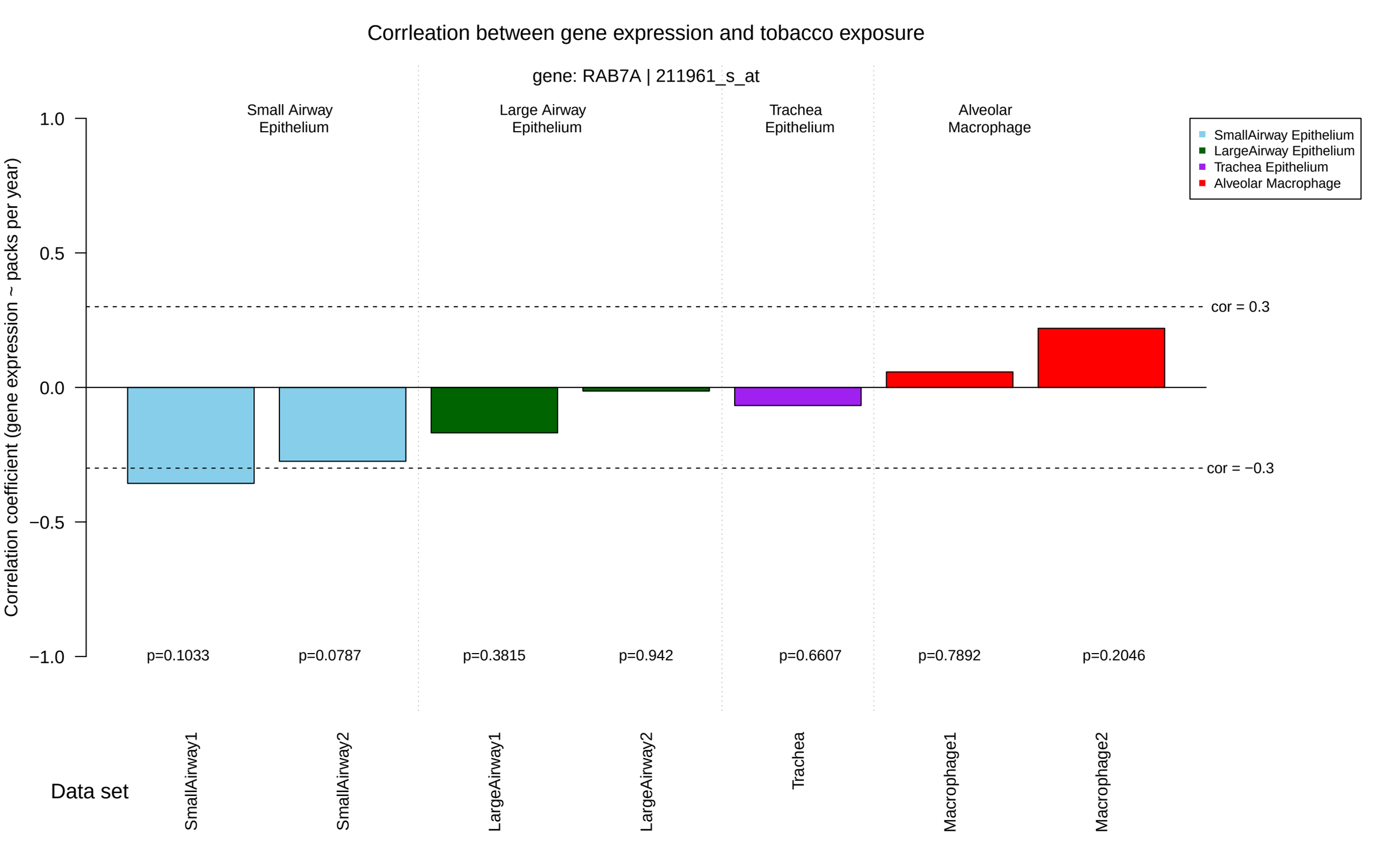

set: LargeAirway1 (p=0.0709)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6333)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4905)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2319)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0092)
RAB7A | 211961_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB7A | 211961_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1033
p=0.0787
p=0.3815
p=0.942
p=0.6607
p=0.7892
p=0.2046
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2